

My Design: Water Filter

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Element C: Presentation and Justification of Solution Design Requirements

Priority	Design Requirements	Explanation for priority	Meet need of stakeholders	What is measured
1 (high priority)	More storage space and improved organization	The teachers need more space to store their files and daily plans so that the area that these files are currently at can be freed up for more sensory devices to be installed in place of the shelves more focused on being a sensory room.	-Helps teachers and school staff with organization and efficiency. -Helps students as organization has been proven to help with sensory overload in kids with autism.	The volume of space the island will be able to be used to accommodate their files.
2 (high priority)	Appealing tactiles	Integrating the sensory devices into the design of even the mundane things will help keep a consistent design focus throughout the space.	-The more appealing the tactiles are, the more the kids will enjoy using them, helping the staff keep the kids focused. -Parents may notice a positive change in the behavior of their child	How often the kids go to use the wall and if it is preferred to use it over other similar sensory devices.
3 (high priority)	Interchangeable	The kids' preferences change and so being able to alter the space or its sensory devices are important to ensuring the project's usefulness.	-Kids will be able to change out tactiles to fit their desires. -Teachers will be able to change out items to	How costly it is to make different variations and how easily they are swapped.

			accommodate changes over time.	
4 (high priority)	Accessibility	The doors and the covers of the cubbies need to be easily opened/removed for the teachers to be able to access the files in it, but needs to be complicated enough for the kids not to open the doors. Files and items must be easily accessed by the staff while in storage, while also remaining hidden and restricted from the children.	-Teachers can access files and other items kept in storage. -Kids will be kept from items they shouldn't be touching. -Fire marshall will approve as all areas can be easily accessed.	Can the kids access things they shouldn't have access to? Can the staff access all files and items they may need to access?
5 (high priority)	Cost	It creates a limit on the quality assurance of the designs.	-Quality affected by cost will impact the children and staff operating in the sensory room	Staying within budget.
6 (low priority)	Durability	Selecting the correct material to make sure the infrastructure doesn't wear out and suit enough for the functions that we want to get from the wood.	-Teachers/staff will benefit from high quality material as there will be less need for future adjustments and repairs.	They withstand any misuse or mistreatment of the designs by the kids.
7 (low priority)	Uniform design	Changes made to room should be part of consistent theme and fit well with the environment.	-Since this is a form of organization, the kids will benefit from reduced sensory overload -Aesthetically pleasing to visitors or users of	Does the sensory space have a consistency in its organization and layout?

			any sort.	
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Priorities:

1. More storage space and improved organization (high priority)
2. Appealing tactiles (high priority)
3. Interchangeable (high priority)
4. Accessibility (high priority)
5. Cost (high priority)
6. Durability (low priority)
7. Uniform design (low priority)

Stakeholder Validation:

After talking with Ms. Brennan, we were given validation for our priorities under the condition that we don't start something we can't finish. To do this, we need to make sure we have an in-depth timeline that our stakeholders are aware of and in agreement with. Because the space is accessed multiple times a day, it is crucial that we carefully consider a manufacturing plan. Part of this is consideration of completing the project in phases, and keeping the main goal of organizing in mind. This may mean saving tactile additions for another project.